SIMATS ENGINEERING

ASSESSMENT TOOL 1 – CASE STUDY

Course Name: Artificial Intelligence
Course Code: CSA1717
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Assessment Title: Case Study

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1. CASE STUDY 1 – SMART MEDICAL DIAGNOSIS SYSTEM

1.1 Problem Statement

A hospital uses a rule-based AI system to identify possible diseases from patient symptoms. The system applies Propositional Logic, Forward Chaining, and Backward Chaining.

Patient A facts:

- Fever = True
- Cough = True
- Breathlessness = True

1.2 Objective

1. Represent medical knowledge using Propositional Logic.
2. Apply Forward Chaining to derive diagnoses.

3. Apply Backward Chaining to verify Pneumonia.

1.3 Knowledge Representation

Let:

- **F** = Fever
- **C** = Cough
- **R** = Rash
- **B** = Breathlessness
- **Flu** = Possible Flu
- **Measles** = Possible Measles
- **Pneumonia** = Possible Pneumonia

Rules

R1: F ∧ C → Flu
R2: F ∧ R → Measles
R3: C ∧ B → Pneumonia

Facts: F, C, B

1.4 Forward Chaining

Step	Rule	Conditions	Derived Fact
0	Facts	F, C, B	Initial facts
1	R1	F ∧ C	Flu
2	R2	F ∧ R	Cannot fire
3	R3	C ∧ B	Pneumonia

Result

Flu = True
Pneumonia = True
Measles = Not Concluded

1.5 Backward Chaining

Goal: Pneumonia

Rule producing Pneumonia:

C ∧ B → Pneumonia

Sub-goals:

Cough → True
Breathlessness → True

Since both conditions are available:

C ∧ B → Pneumonia

Therefore, **Pneumonia is successfully proved.**

1.6 Analysis and Conclusion

The system successfully derives Flu and Pneumonia using Forward Chaining. Backward Chaining verifies Pneumonia because Cough and Breathlessness are known facts. This demonstrates how logical reasoning can support preliminary decision-making.

2. CASE STUDY 2 – AUTONOMOUS TRAFFIC MANAGEMENT AGENT

2.1 Problem Statement

A smart-city traffic system uses First-Order Logic to provide priority to emergency vehicles and determine whether vehicles following them can proceed.

2.2 Knowledge Base

Rules

R1: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)

R2: ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x)

R3: Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x)

Facts

- Vehicle(Ambulance)
- EmergencyType(Ambulance)
- Vehicle(CarA)
- Behind(CarA, Ambulance)

2.3 Unification

Match Rule 1 with the ambulance facts.

Vehicle(x) with **Vehicle(Ambulance)**

Therefore:

θ = {x / Ambulance}

After substitution:

Vehicle(Ambulance) ∧ EmergencyType(Ambulance)
→ ClearPath(Ambulance)

Hence:

ClearPath(Ambulance)

Using Rule 2:

GreenSignal(Ambulance)

Now Rule 3 gives:

Vehicle(CarA) ∧ Behind(CarA, Ambulance) ∧ GreenSignal(Ambulance)
→ **Proceed(CarA)**

Therefore, **Proceed(CarA) is true.**

2.4 Resolution

CNF Forms

R1:

¬Vehicle(x) ∨ ¬EmergencyType(x) ∨ ClearPath(x)

R2:

¬ClearPath(x) ∨ GreenSignal(x)

R3:

¬Vehicle(x) ∨ ¬Behind(x,y) ∨ ¬GreenSignal(y) ∨ Proceed(x)

Negate the goal:

¬Proceed(CarA)

Using the facts and rules:

ClearPath(Ambulance)
↓
GreenSignal(Ambulance)
↓
Proceed(CarA)

Together with:

¬Proceed(CarA)

we obtain:

Proceed(CarA) ∧ ¬Proceed(CarA) → □

The empty clause □ proves the goal.

2.5 Multiple Emergency Vehicles

The current rules can derive clear paths for multiple emergency vehicles if their facts are available. However, additional rules are required for:

- Priority between emergency vehicles
- Route conflicts
- Green-signal scheduling
- Vehicle location and direction

Example:

EmergencyVehicle(x) ∧ HigherPriority(x,y) → GivePriority(x)

2.6 Conclusion

FOL, Unification, and Resolution successfully prove that **CarA can proceed**. Additional priority and routing rules are required for handling multiple simultaneous emergencies safely.

3. CASE STUDY 3 – AGRICULTURAL EXPERT REASONING SYSTEM

3.1 Problem Statement

An agricultural AI system uses propositional logic to determine irrigation requirements and crop risk.

3.2 Knowledge Base

Let:

- **S** = SoilDry
- **I** = IrrigationNeeded
- **W** = CropWheat
- **D** = ApplyDripMethod
- **R** = CropAtRisk

Rules

P1: $S \rightarrow I$

P2: $I \land W \rightarrow D$

P3: $\neg D \land W \rightarrow R$

Facts

S, W

3.3 CNF Conversion

P1

$S \rightarrow I$

$\boldsymbol{\neg S \lor I}$

P2

$I \land W \rightarrow D$

$\boldsymbol{\neg I \lor \neg W \lor D}$

P3

$\neg D \land W \rightarrow R$

$\boldsymbol{D \lor \neg W \lor R}$

Complete CNF

- C1: $\neg S \lor I$

- C2: ¬I ∨ ¬W ∨ D
- C3: D ∨ ¬W ∨ R
- C4: S
- C5: W

3.4 Resolution Refutation

Goal: D

Negated goal:

¬D

Step 1

C1 + C4:

¬S ∨ I
S

Therefore:

I

Step 2

C2 + I:

¬I ∨ ¬W ∨ D
I

Therefore:

¬W ∨ D

Step 3

C5 + ¬W ∨ D:

W
¬W ∨ D

Therefore:

D

Step 4

D + ¬D:

□

Therefore:

ApplyDripMethod is proved.

3.5 Removing SoilDry

If **S** is removed:

- IrrigationNeeded cannot be derived.
- ApplyDripMethod cannot be derived.
- CropAtRisk also cannot be concluded.

This is because absence of **D** does not mean **¬D**.

3.6 Conclusion

The Resolution Refutation proves that the drip irrigation method should be applied when SoilDry and CropWheat are known. Removing SoilDry breaks the inference chain.

4. CASE STUDY 4 – WUMPUS WORLD KNOWLEDGE AGENT

4.1 Problem Statement

Wumpus World demonstrates logical reasoning under incomplete information. The agent receives:

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

4.2 Knowledge Representation

Rules

R1: Stench[1,2] → Wumpus adjacent to [1,2]

R2: Breeze[1,1] → Pit adjacent to [1,1]

R3: Glitter[x,y] → Gold[x,y]

R4: ¬Wumpus[x,y] ∧ ¬Pit[x,y] → Safe[x,y]

4.3 Belief State and Modus Ponens

Step 1

Stench[1,2]

Therefore:

Wumpus is adjacent to [1,2]

Step 2

Breeze[1,1]

Therefore:

Pit is adjacent to [1,1]

Step 3

Glitter[2,2]

Therefore:

Gold[2,2]

4.4 Forward Chaining

Possible Wumpus Locations

Adjacent cells to [1,2]:

- [1,1]
- [1,3]
- [2,2]

Therefore, possible Wumpus locations are:

[1,1], [1,3], [2,2]

Possible Pit Locations

Adjacent cells to [1,1]:

- [1,2]
- [2,1]

Therefore, possible Pit locations are:

[1,2], [2,1]

The exact locations cannot be determined.

4.5 Path Safety Analysis

Required path:

[1,1] → [1,2] → [2,2]

The Glitter proves:

Gold[2,2]

However, there is no explicit information proving:

¬Wumpus[2,2] and **¬Pit[2,2]**

Therefore, Rule 4 cannot prove **Safe[2,2]**.

Decision

The path **cannot be guaranteed to be safe** with the available knowledge.

4.6 Conclusion

The agent successfully identifies the gold at [2,2] and possible hazard locations. However, incomplete information prevents it from proving that the entire path is safe.

6. REFERENCES

1. S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed., Pearson, 2021.

2. W. Ertel, *Introduction to Artificial Intelligence*, 3rd ed., Springer, 2023.
3. M. Wooldridge, *The Road to Conscious Machines: The Story of AI*, Penguin Books, 2021.
4. I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press.